5 $G(S) = \frac{10+12}{S(S+2)}$



$H(s) = 1$

a) 1) System type: 1

2) $k_p = G(0) = \frac{22}{0(0+2)} = \infty$

$e_{ss} = \frac{1}{1+k_p} = \boxed{0}$

3) $k_v = \lim_{S \to 0} S\,G(S) = \frac{22S}{S(S+2)}\Big|_{S=0} = \frac{22}{0+2} = 11$

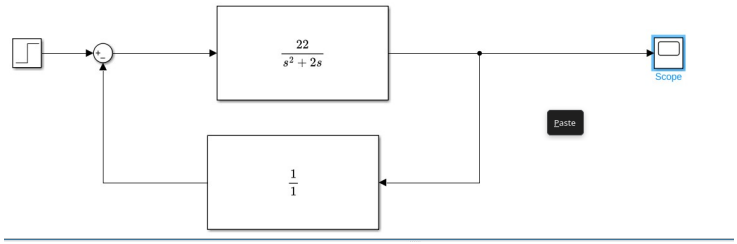
$e_{ss} = \frac{1}{k_v} = \frac{1}{11} = \boxed{0.09}$ percentage is 9%

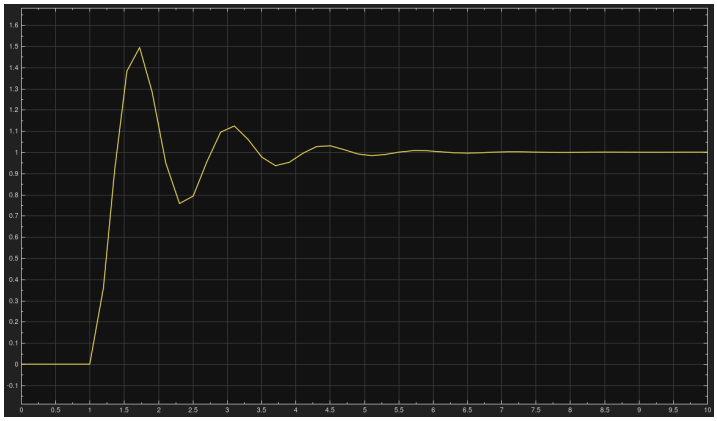
4) $k_a = \lim_{S \to 0} S^2 G(S) = \frac{22S^2}{S(S+2)}\Big|_{S=0} = \boxed{0}$

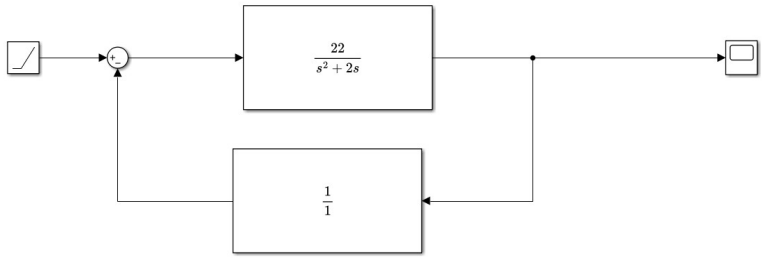
$e_{ss} = 1/k_a = 1/0 = \boxed{\infty}$

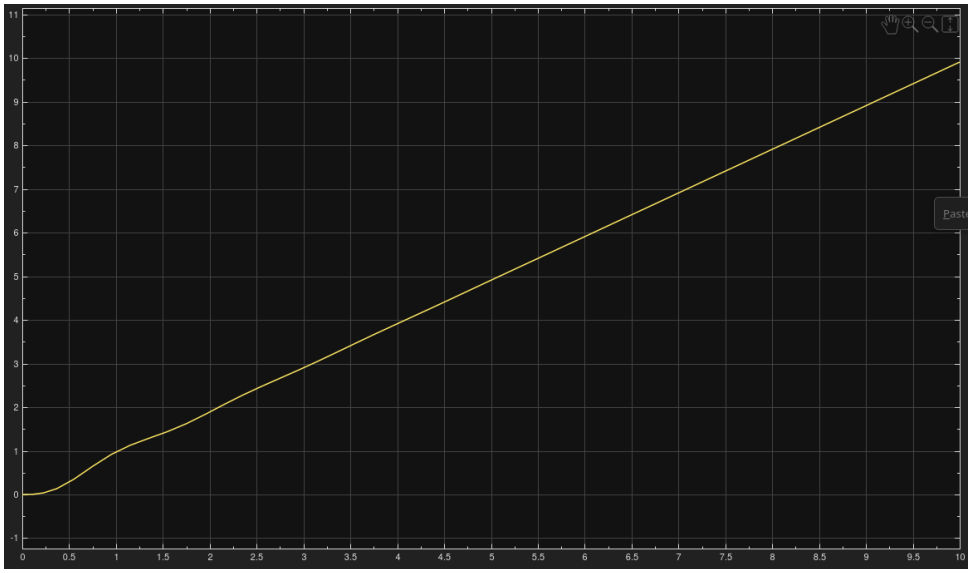
b)

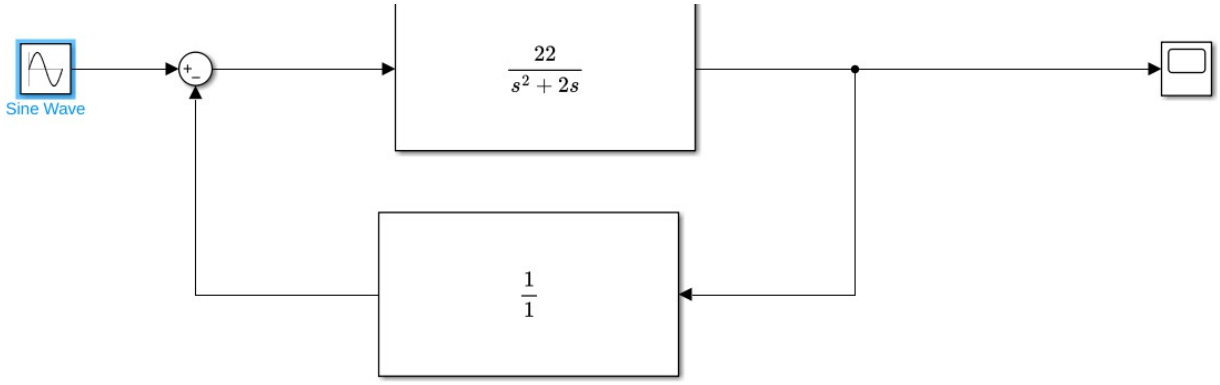
5.

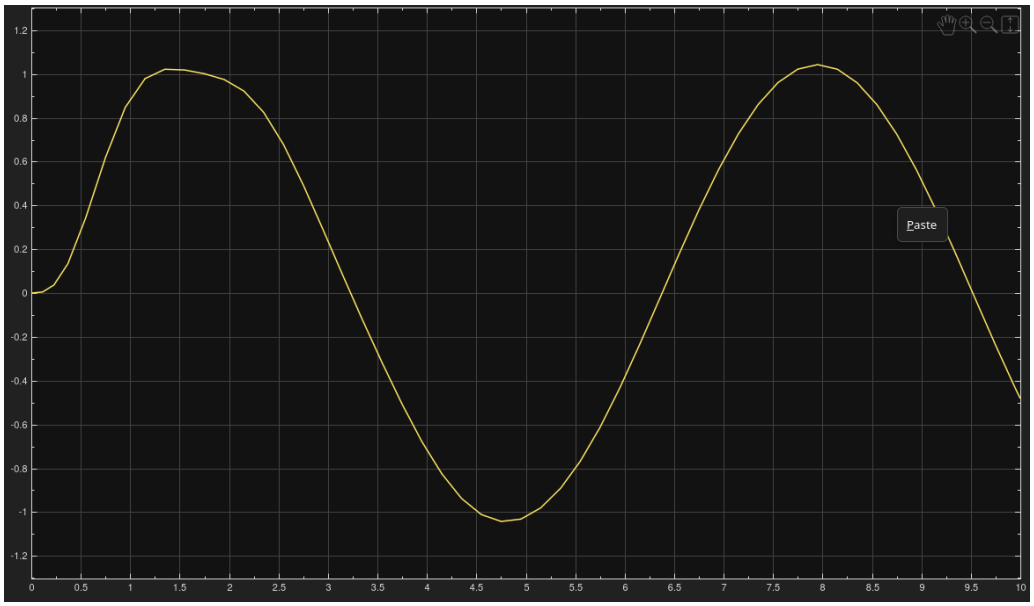









Sine Wave
$\frac{22}{s^2+2s}$
$\frac{1}{1}$


1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6
6.5
7
7.5
8
8.5
9
9.5
10
Paste